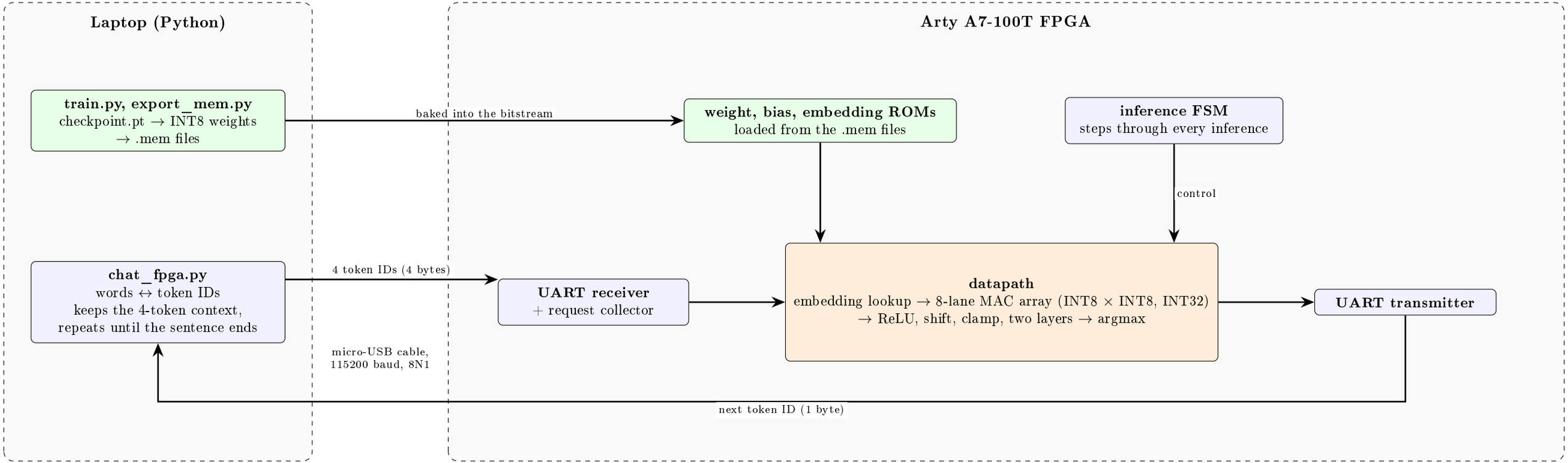
Laptop (Python)
Arty A7-100T FPGA
train.py, export_mem.py
checkpoint.pt → INT8 weights
→ .mem files
baked into the bitstream
weight, bias, embedding ROMs
loaded from the .mem files
inference FSM
steps through every inference
control
chat_fpga.py
words ↔ token IDs
keeps the 4-token context,
repeats until the sentence ends
4 token IDs (4 bytes)
UART receiver
+ request collector
datapath
embedding lookup → 8-lane MAC array (INT8 × INT8, INT32)
→ ReLU, shift, clamp, two layers → argmax
UART transmitter
micro-USB cable,
115200 baud, 8N1
next token ID (1 byte)